

Major Project - Placement Management System (MySQL Only)

- Below are the Sample data for the project needed and Relation Diagram,
- After this you can see the task wise project making questions
- Complete the project and upload all the commands into Github and Submit it.

1. Departments Table

dept_id	dept_name
101	CSE
102	ECE
103	EEE
104	Mechanical
105	Civil

2. Students Table

student_id	student_name	email	phone	cgpa	dept_id
1	Saran	saran@gmail.com ↗	9876543210	8.5	101
2	Priya	priya@gmail.com ↗	9876543211	9.0	102
3	Rahul	rahul@gmail.com ↗	9876543212	7.8	101
4	Deepa	deepa@gmail.com ↗	9876543213	8.9	103
5	Anjali	anjali@gmail.com ↗	9876543214	8.2	102
6	Kumar	kumar@gmail.com ↗	9876543215	7.5	104
7	Sneha	sneha@gmail.com ↗	9876543216	9.1	101
8	Varun	varun@gmail.com ↗	9876543217	8.0	105
9	Charan	charan@gmail.com ↗	9876543218	7.2	104
10	Harsha	harsha@gmail.com ↗	9876543219	8.7	101

3. Companies Table

company_id	company_name	location	package_lpa
1	TCS	Hyderabad	4.5
2	Infosys	Bangalore	5.2
3	Accenture	Chennai	6.0
4	Wipro	Pune	4.2
5	Cognizant	Hyderabad	5.8

4. Jobs Table

job_id	company_id	role_name	min_cgpa
1	1	Java Developer	7.5
2	2	Python Developer	8.0
3	3	Full Stack Developer	8.2
4	4	Data Analyst	7.0
5	5	Software Engineer	7.8

5. Applications Table

application_id	student_id	job_id	status
1	1	1	Applied
2	1	5	Applied
3	2	3	Applied
4	3	1	Applied
5	4	2	Applied
6	5	3	Applied
7	6	4	Applied
8	7	5	Applied
9	8	2	Applied
10	10	3	Applied

6. Aptitude Tests Table

test_id	student_id	marks
1	1	85
2	2	92
3	3	70
4	4	95
5	5	80
6	6	65
7	7	98
8	8	75
9	9	68
10	10	88

7. Interviews Table

interview_id	student_id	company_id	round_name	result
1	1	1	Technical	Pass
2	2	3	Technical	Pass
3	3	1	Technical	Fail
4	4	2	Technical	Pass
5	5	5	Technical	Pass
6	6	4	Technical	Fail
7	7	5	Technical	Pass
8	8	2	Technical	Pass
9	10	3	Technical	Pass

8. Placements Table

placement_id	student_id	company_id	package_lpa
1	1	1	4.5
2	2	3	6.0
3	4	2	5.2
4	5	5	5.8
5	7	5	5.8
6	8	2	5.2
7	10	3	6.0

9. Skills Table

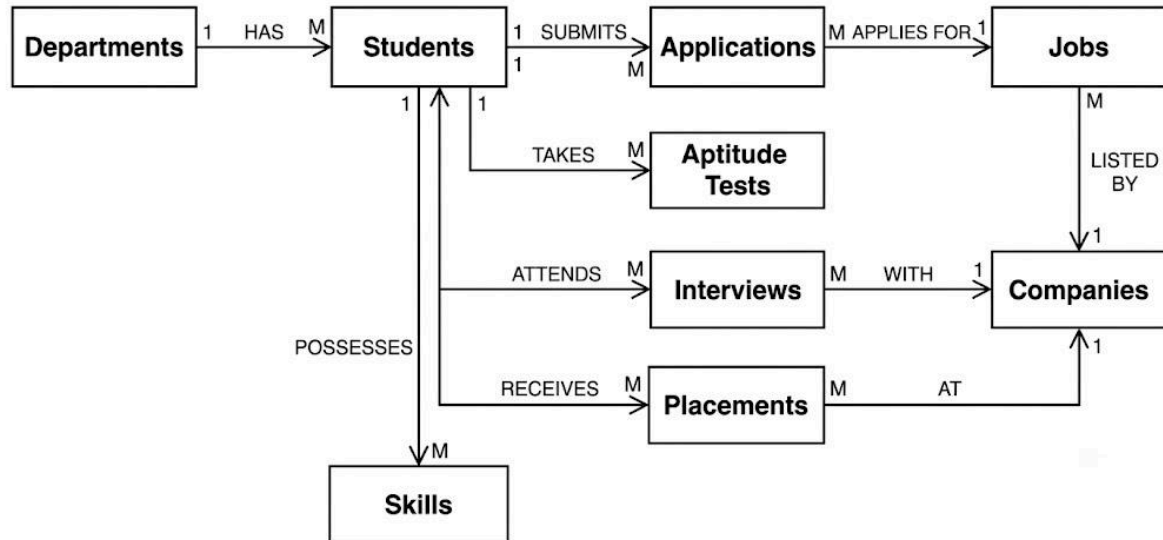
skill_id	skill_name
1	Java
2	Python
3	SQL
4	React
5	Spring Boot

10. Student Skills Table

student_id	skill_id
1	1
1	3
2	2
2	4
3	1
4	2
5	5
7	1
7	5
10	3

Relationship Diagram

UNIVERSITY PLACEMENT SYSTEM ER DIAGRAM



Module 1: Department Management

Task 1

Create a table named **departments** with the following attributes:

- dept_id
- dept_name

Ensure:

- dept_id is the PRIMARY KEY.
- dept_name is UNIQUE and NOT NULL.

Task 2

Insert at least 5 departments into the table.

Example:

- CSE
- ECE

- EEE
- Mechanical
- Civil

Module 2: Student Management

Task 3

Create a table named **students** with the following attributes:

- student_id
- student_name
- email
- phone
- cgpa
- dept_id

Ensure:

- student_id is AUTO_INCREMENT and PRIMARY KEY.
- email is UNIQUE.
- student_name cannot be NULL.
- dept_id should reference departments table.

Task 4

Insert at least 10 students belonging to different departments.

Task 5

Display all student records.

Module 3: Company Management

Task 6

Create a table named **companies** with the following attributes:

- company_id
- company_name
- location
- package_lpa

Ensure:

- company_id is PRIMARY KEY.
- company_name is UNIQUE.

Task 7

Insert at least 5 companies.

Examples:

- TCS
- Infosys
- Accenture
- Wipro
- Cognizant

Task 8

Display all companies.

Module 4: Job Openings

Task 9

Create a table named **jobs** with the following attributes:

- job_id
- company_id
- role_name
- min_cgpa

Establish a relationship between jobs and companies.

Task 10

Insert at least 5 job openings.

Examples:

- Java Developer
- Python Developer
- Full Stack Developer
- Data Analyst
- Software Engineer

Task 11

Display all job openings.

Module 5: Student Applications

Task 12

Create a table named **applications** with the following attributes:

- application_id
- student_id
- job_id
- status

Create relationships with:

- students table
- jobs table

Task 13

Insert application records for students.

Task 14

Display all applications.

Module 6: Aptitude Test

Task 15

Create a table named **aptitude_tests** containing:

- test_id
- student_id
- marks

Create a relationship with students table.

Task 16

Insert aptitude marks for all students.

Task 17

Display aptitude test results.

Module 7: Interview Management

Task 18

Create a table named **interviews** containing:

- interview_id
- student_id
- company_id
- round_name
- result

Create relationships with:

- students
- companies

Task 19

Insert interview details for students.

Task 20

Display all interview records.

Module 8: Placement Management

Task 21

Create a table named **placements** containing:

- placement_id
- student_id
- company_id
- package_lpa

Create relationships with:

- students
- companies

Task 22

Insert placement records.

Task 23

Display all placed students.

DQL Tasks

Task 24

Display student names along with department names.

Task 25

Display student names and the companies they applied for.

Task 26

Display all students whose CGPA is greater than 8.

Task 27

Display companies offering packages greater than 5 LPA.

Task 28

Display students who attended interviews.

Task 29

Display students who got placed.

Task 30

Display students who are not yet placed.

Aggregate Functions

Task 31

Find the total number of students.

Task 32

Find the average CGPA of students.

Task 33

Find the highest CGPA.

Task 34

Find the lowest CGPA.

Task 35

Display department-wise student count.

Task 36

Display company-wise placed student count.

GROUP BY and HAVING**Task 37**

Display departments having more than 2 students.

Task 38

Display companies that selected more than one student.

Task 39

Display departments whose average CGPA is greater than 8.

JOIN Tasks**Task 40**

Display student names and department names using INNER JOIN.

Task 41

Display student names and company names of applied jobs.

Task 42

Display all companies and their jobs.

Task 43

Display all students and their applications.

Task 44

Display students who have not applied to any job.

Subquery Tasks

Task 45

Display the student having the highest CGPA.

Task 46

Display students whose CGPA is above average.

Task 47

Display companies offering the maximum package.

Task 48

Display students who are placed in TCS.

Task 49

Display students who applied but were not selected.

View Tasks

Task 50

Create a view named **selected_students** containing:

- student_name
- company_name
- package_lpa

Task 51

Display records from the view.

Task 52

Create a view for students having CGPA greater than 8.

Stored Procedure Tasks

Task 53

Create a stored procedure to display all students.

Task 54

Create a stored procedure that accepts a department ID and displays students of that department.

Task 55

Create a stored procedure to display placed students.

Task 56

Create a procedure to insert a new student.

Task 57

Create a procedure to update a student's CGPA.

Trigger Tasks

Task 58

Create a trigger that automatically sets application status as "Applied" before insertion.

Task 59

Create a trigger to prevent insertion of students whose CGPA is below 6.

Task 60

Create a trigger that stores deleted student records into a backup table.